CS683 PA2: Dynamic I-RRIP

Single-core ChampSim evaluation using Trace1.gz, Trace2.gz and Trace3.gz. Each run uses 25 million warmup instructions and 25 million simulation instructions.

Methodology

Set dueling: L2 sets 0-15 are fixed SRRIP leaders, sets 16-31 are fixed I-RRIP leaders, and all remaining sets are followers controlled by the selector.

Selector statistic: Counters are updated on accesses to either leader group. Let I_A and I_M be instruction accesses and misses, and A and M be total accesses and misses in leader sets. The score is **(I_A + I_M) / (A + M)**. The selector chooses I-RRIP for followers when the score exceeds the configured threshold; otherwise it chooses SRRIP. The tested thresholds are 0.10, 0.25, 0.50, 0.75 and 0.90.

Insertion and replacement: Hits reset RRPV to 0. Instruction lines are inserted at RRPV 2. Under I-RRIP, data lines are inserted at RRPV 3; otherwise lines are inserted at RRPV 2. Victim selection chooses a line at RRPV 3, incrementing RRPVs when needed.

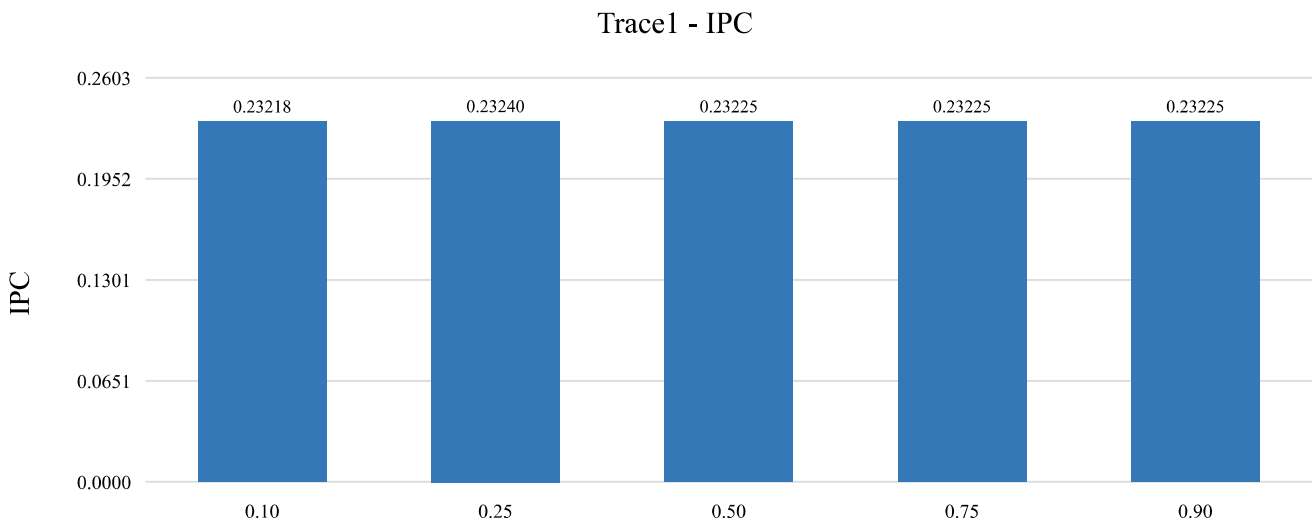
Measurement: Each trace is evaluated independently with identical warmup and simulation lengths. Speedup is configuration IPC divided by that trace's baseline SRRIP IPC.

Results

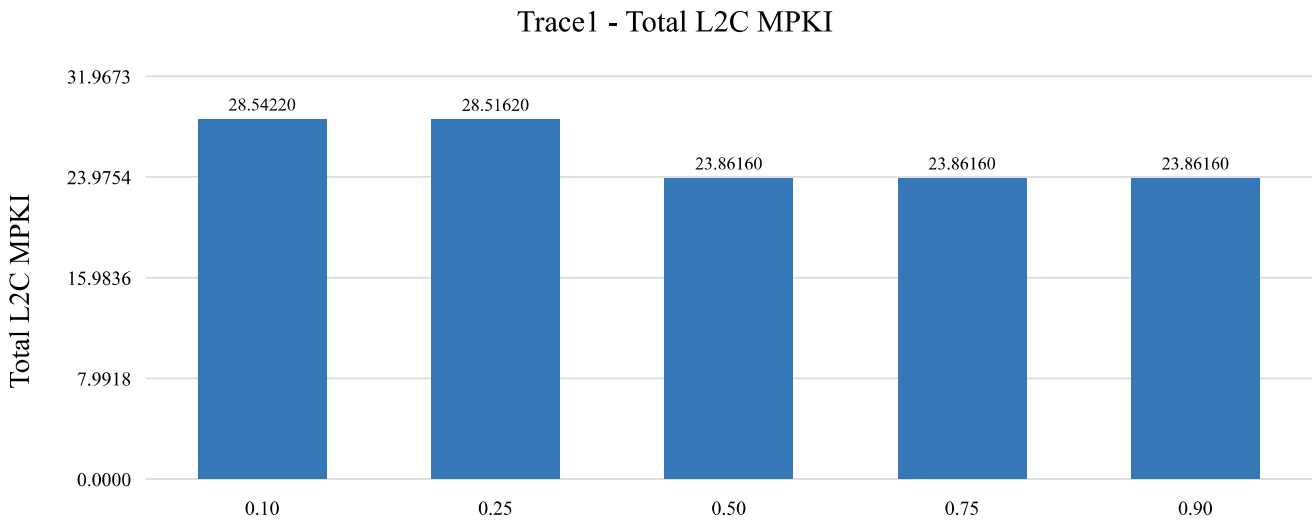
Trace1

Threshold	IPC	Speedup	Total MPKI	Data-load MPKI	Instruction-load MPKI	Follower policy
baseline	0.232232	1.000000x	23.79910	7.06972	10.52550	SRRIP
0.10	0.232178	0.999767x	28.54220	8.67768	9.66028	I-RRIP
0.25	0.232396	1.000706x	28.51620	8.65824	9.65660	I-RRIP
0.50	0.232250	1.000078x	23.86160	7.09548	10.50980	RRIP
0.75	0.232250	1.000078x	23.86160	7.09548	10.50980	RRIP
0.90	0.232250	1.000078x	23.86160	7.09548	10.50980	RRIP

Trace1: IPC by threshold



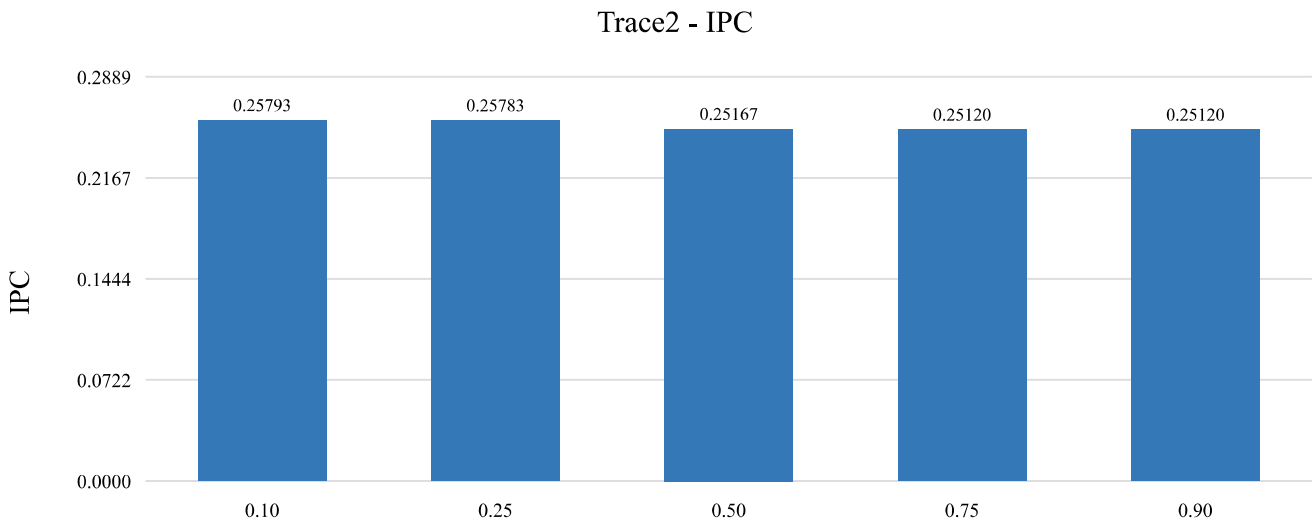
Trace1: Total MPKI by threshold



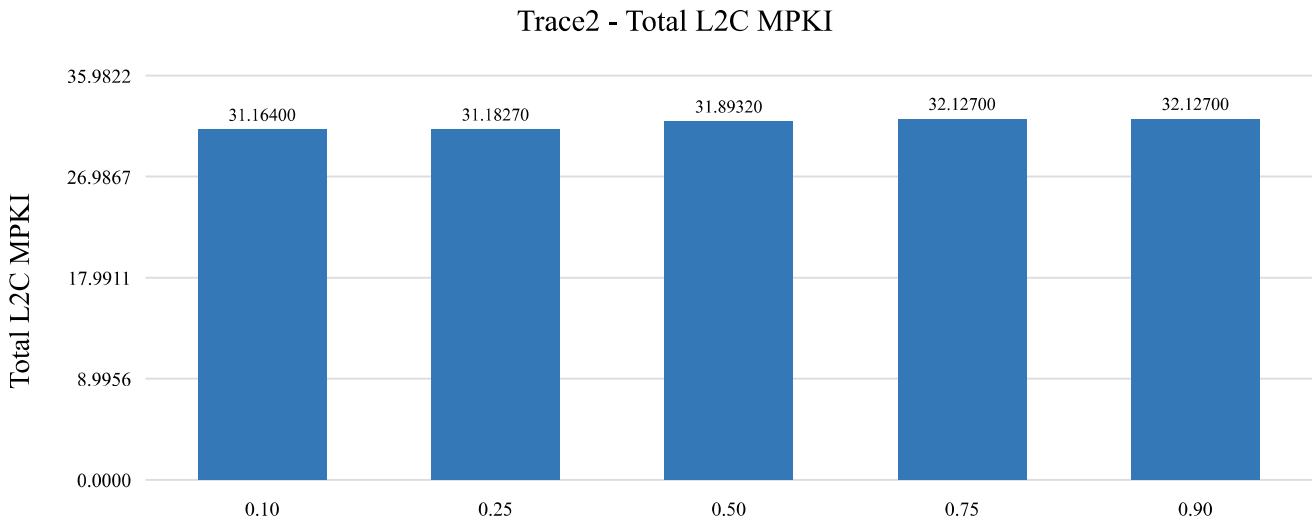
Trace2

Threshold	IPC	Speedup	Total MPKI	Data-load MPKI	Instruction-load MPKI	Follower policy
baseline	0.251085	1.000000x	32.09980	7.99564	18.21980	SRRIP
0.10	0.257929	1.027258x	31.16400	9.80224	11.83680	I-RRIP
0.25	0.257830	1.026863x	31.18270	9.80504	11.85890	I-RRIP
0.50	0.251670	1.002330x	31.89320	8.00920	17.94300	RRIP
0.75	0.251202	1.000466x	32.12700	8.02352	18.16160	RRIP
0.90	0.251202	1.000466x	32.12700	8.02352	18.16160	RRIP

Trace2: IPC by threshold



Trace2: Total MPKI by threshold

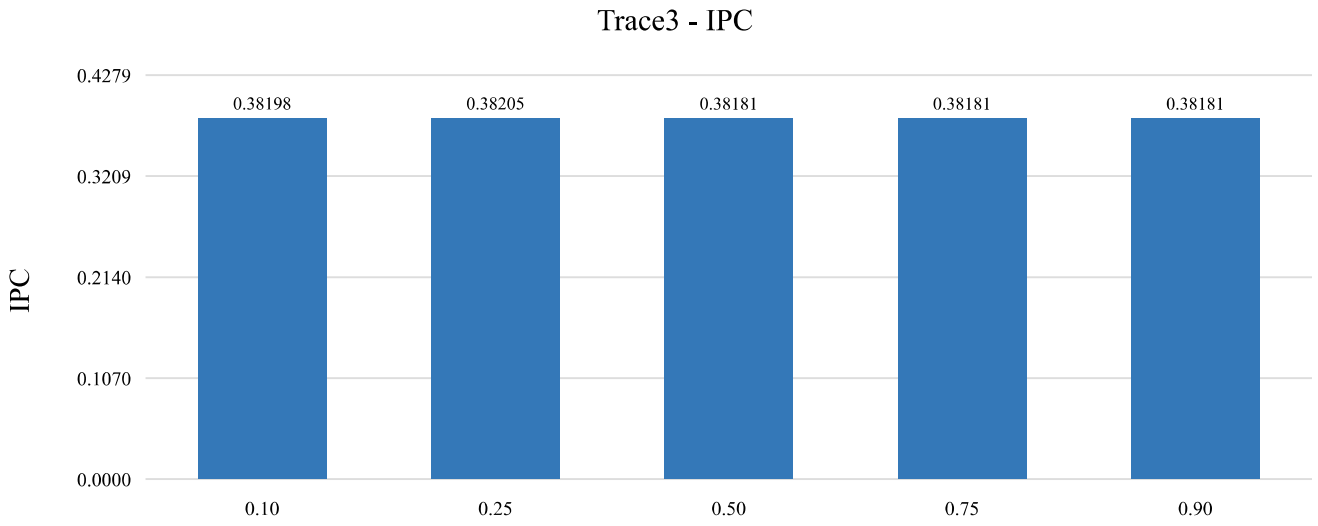


Trace3

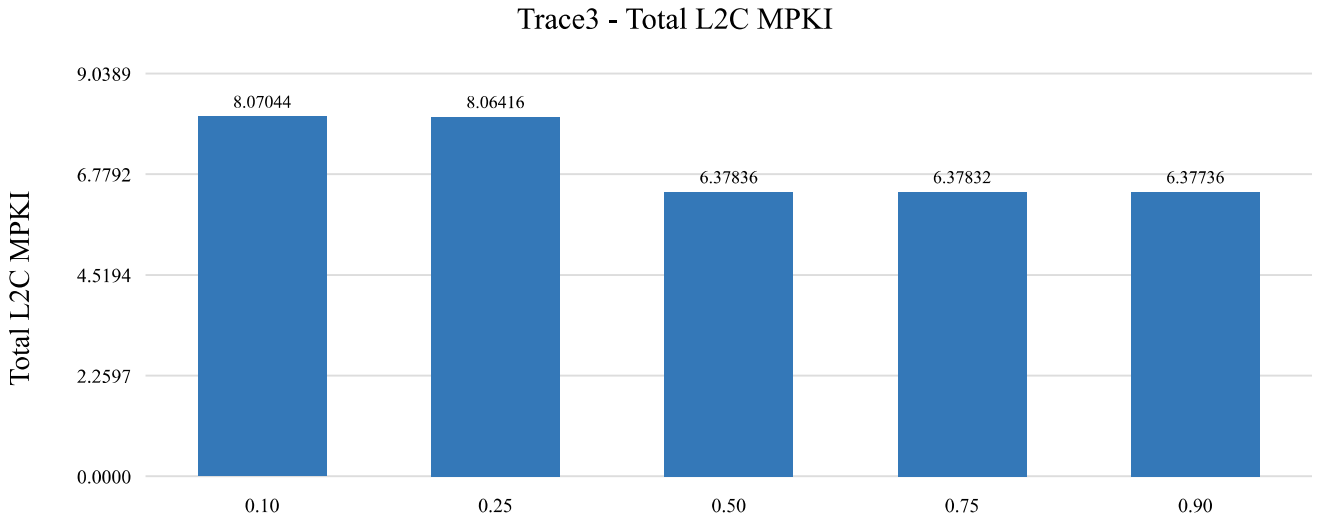
Threshold	IPC	Speedup	Total MPKI	Data-load MPKI	Instruction-load MPKI	Follower policy
baseline	0.381789	1.000000x	6.34360	2.84660	1.19336	SRRIP
0.10	0.381975	1.000487x	8.07044	3.06984	0.95636	I-RRIP
0.25	0.382055	1.000697x	8.06416	3.06912	0.95468	I-RRIP
0.50	0.381808	1.000050x	6.37836	2.85188	1.18996	RRIP

0.75	0.381808	1.000050x	6.37832	2.85188	1.18996	RRIP
0.90	0.381805	1.000042x	6.37736	2.85140	1.18952	RRIP

Trace3: IPC by threshold



Trace3: Total MPKI by threshold



Discussion

Trace1: Baseline IPC is 0.232232. The highest measured IPC is 0.232396 at threshold 0.25 (+0.071% versus baseline). Total MPKI changes from 23.7991 to 28.5162. These results show that the effect of the threshold depends on the workload.

Trace2: Baseline IPC is 0.251085. The highest measured IPC is 0.257929 at threshold 0.10 (+2.726% versus baseline). Total MPKI changes from 32.0998 to 31.1640. These results show that the effect of the threshold depends on the workload.

Trace3: Baseline IPC is 0.381789. The highest measured IPC is 0.382055 at threshold 0.25 (+0.070% versus baseline). Total MPKI changes from 6.3436 to 8.0642. These results show that the effect of the threshold depends on the workload.